

Project Overview- Supply Chain Delivery Prediction

The project is designed to predict late delivery risks in a supply chain dataset using machine learning. The project steps are summarized as follows:

1. Data Cleaning

- Removed missing values from the dataset to ensure accuracy.
- Verified the consistency and validity of all features.

2. Data Processing (Encoding)

- Converted categorical features (Shipping_Mode, Order_Region) into numerical values using Label Encoding.
- Created a new feature Delay to represent the difference between actual and expected shipping days.

3. Machine Learning Models

- Built two models: Random Forest Classifier and Neural Network (MLPClassifier).
- Selected important features such as Shipping_Mode, Order_Region, Order_Item_Quantity, Product_Price, Shipping_Days, and Delay.

4. Model Training

- Split the dataset into training (80%) and testing (20%) sets.
- Performed feature scaling for the Neural Network model.
- Applied GridSearchCV to find the best hyperparameters for both models.

5. Model Evaluation

- Evaluated models using Accuracy, Mean Squared Error, Classification Report, and Confusion Matrix.
- Visualized Feature Importance and Confusion Matrices to interpret results.

6. Results Interpretation

- Identified Delay as the most important feature affecting late delivery risk.
- Random Forest provides interpretable feature importance, while Neural Network captures complex patterns.

7. Report Writing

- Documented all steps from data preprocessing to model evaluation.
- Provided visualizations and explanations to support project insights.

Overall, this project demonstrates how data-driven techniques can help a business **predict shipment delays**, optimize supply chain operations, and improve customer satisfaction.